

Supplementary Material — Expert–algorithm agreement

Supplementary — Expert– automated morphological agreement (n = 54)

Ordinal order for quadratic weighted κ : Dead tree $\rightarrow$ Tree in bud $\rightarrow$ Glomerular $\rightarrow$ Seafan $\rightarrow$ Medusa. Bootstrap 95% CI: 10 000 resamples, seed 20260727.

- Overall agreement: **57.4%** (31/54)
- Quadratic weighted κ : **0.507** (95% CI **0.222–0.714**)

Confusion matrix (rows = expert, columns = automated)

expert automated	Dead tree	Tree in bud	Glomerular	Seafan	Medusa
Dead tree	2	0	0	1	0
Tree in bud	0	2	3	0	0
Glomerular	0	0	20	7	6
Seafan	0	0	3	0	0
Medusa	0	0	3	0	7

Source: `grading/agreement_stats.md` / `grading/confusion_matrix.csv`.